

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · COMBINED SCIENCE

COMBINED SCIENCE 5006/1

PAPER 1 November 2022

1 hour

PAPER 1 — MULTIPLE CHOICE — 40 QUESTIONS — 1 HOUR

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to November 2022 5006/1.

Additional materials: Answer all 40 questions. For each question there are four possible answers A, B, C and D.

INSTRUCTIONS: Choose the one you consider correct and record it. Each question carries 1 mark.

1. Oxygenated blood is pumped to the body from the **[1]**

- A) right ventricle
- B) left ventricle
- C) pulmonary vein only, without the heart
- D) right atrium

2. Which gas is produced during photosynthesis? **[1]**

- A) hydrogen
- B) oxygen
- C) nitrogen
- D) methane

3. Malaria is caused by a pathogen transmitted by **[1]**

- A) a fungus by contact
- B) Plasmodium, via the Anopheles mosquito
- C) a virus in air
- D) a bacterium in contaminated water

4. Anaerobic respiration in yeast produces **[1]**

- A) starch
- B) oxygen
- C) ethanol and carbon dioxide
- D) lactic acid only

5. Friction is a force that **[1]**

- A) speeds objects up in all cases
- B) is measured in kg
- C) has no direction
- D) opposes motion between surfaces

6. A human sperm cell has a flagellum so that it can **[1]**

- A) swim towards the egg
- B) photosynthesise
- C) carry oxygen
- D) store yolk

7. Electrolysis of aqueous copper(II) sulfate with copper electrodes is used to [1]

- A) liquefy air
- B) make sodium
- C) extract aluminium
- D) purify copper

8. Alkali metals (Group 1) become more reactive [1]

- A) only in period 2
- B) when they lose the nucleus
- C) down the group
- D) up the group

9. Convection occurs in [1]

- A) fluids (liquids and gases)
- B) solids only
- C) crystals only
- D) a vacuum

10. In double circulation, blood passes through the heart [1]

- A) not at all
- B) only at night
- C) twice per circuit of the body
- D) once per circuit of the body

11. The pitch of a sound depends on its [1]

- A) loudness only
- B) frequency
- C) colour
- D) amplitude

12. Universal indicator in a strong alkali is [1]

- A) red
- B) colourless always
- C) green
- D) purple / blue

13. Sodium chloride is held together by [1]

- A) covalent bonds only
- B) ionic bonds
- C) hydrogen bonds only
- D) metallic bonds only

14. The SI unit of energy is the [1]

- A) watt
- B) newton
- C) joule
- D) pascal

15. The gravitational force on an object is its [1]

- A) weight
- B) mass
- C) density
- D) volume

16. The most abundant gas in clean dry air is **[1]**

- A) nitrogen
- B) oxygen
- C) argon
- D) carbon dioxide

17. Protease enzymes digest proteins to **[1]**

- A) fatty acids
- B) amino acids
- C) glucose
- D) maltose

18. Eggs (ova) are produced in the **[1]**

- A) testes
- B) ovaries
- C) vagina
- D) bladder

19. Electrons are found in **[1]**

- A) electron shells / energy levels
- B) the vacuole
- C) neutrons only
- D) the nucleus

20. Length is measured with a **[1]**

- A) metre rule / tape
- B) stopwatch
- C) newton meter
- D) thermometer

21. A lighted splint is used to test for hydrogen. A positive result is **[1]**

- A) limewater turning milky
- B) a glowing splint relighting
- C) bleaching litmus
- D) a pop sound

22. Which is a correct order in treating river water for drinking? **[1]**

- A) chlorination → screening
- B) electrolysis → rusting
- C) screening → filtration → chlorination
- D) boiling → mining

23. An acidic solution has **[1]**

- A) pH 14 only
- B) pH greater than 7
- C) pH less than 7
- D) pH 7

24. Which energy conversion is correct? **[1]**

- A) light → mass always
- B) sound → nuclear always
- C) chemical → kinetic in a battery at rest
- D) gravitational potential → kinetic as an object falls

25. The electronic configuration of magnesium (proton number 12) is **[1]**

- A) 2,8,1
- B) 2,6
- C) 2,8,2
- D) 2,8,7

26. During electrolysis, cations move to the **[1]**

- A) anode
- B) fuse
- C) cathode
- D) voltmeter

27. The SI unit of speed is **[1]**

- A) kg
- B) m/s
- C) N
- D) J

28. Time intervals in the laboratory are measured with a **[1]**

- A) stopwatch / timer
- B) ammeter
- C) balance
- D) measuring cylinder

29. Iron rusts when it is in contact with **[1]**

- A) oxygen and water
- B) oil only
- C) nitrogen only
- D) argon

30. Pressure is defined as **[1]**

- A) $F + A$
- B) A / F
- C) $F \times A$
- D) F / A

31. Tuberculosis (TB) is caused by a bacterium and is spread mainly by **[1]**

- A) ticks
- B) droplet infection (coughing, sneezing)
- C) contaminated rivers only
- D) mosquitoes

32. The lungs excrete **[1]**

- A) carbon dioxide
- B) undigested fibre
- C) bile
- D) urea

33. The green pigment that absorbs light for photosynthesis is **[1]**

- A) haemoglobin
- B) melanin
- C) carotene only
- D) chlorophyll

34. Neutral solutions have pH [1]

- A) 14
- B) 0
- C) 7
- D) 1

35. An enzyme is [1]

- A) a carbohydrate that stores energy
- B) a biological catalyst
- C) a type of mineral ion
- D) a waste product of respiration

36. Average speed is [1]

- A) time / distance
- B) distance / time
- C) mass / time
- D) distance × time

37. The lock-and-key model says the substrate fits the [1]

- A) vacuole
- B) nucleus
- C) active site
- D) cell wall

38. In a food chain, a producer is an organism that [1]

- A) eats animals only
- B) breaks down dead matter only
- C) makes its own food by photosynthesis
- D) is always a fungus

39. A dull black surface, compared with a shiny silver surface, [1]

- A) is a good absorber and emitter of radiation
- B) is a poor absorber of radiation
- C) reflects all heat always
- D) cannot emit radiation

40. Which material is magnetic? [1]

- A) iron
- B) wood
- C) plastic
- D) copper

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).